# BIOCHEMISTRY CUME
# APRIL 5, 2002

1. (20pts) Identify the side chain and functional groups in a protein that can form hydrogen bonds or electrostatic bonds (salt bridges) with a lysine side chain at pH 7.

2. (50pts) Draw the side-chain structures of the following pairs of amino acids and explain the differences between them (charge at physiological pH, hydrophobicity, length, size and so on). : Asp and Lys; Thr and Glu; Phe and Leu; Arg and Tyr; Met and Ser.

3. (30pts) Asp102, Ser195 are key active-site residues in chymotrypsin. The negatively charged side-chain carboxyl group of Asp102 and the side chain group of Ser195 take part in the formation of the catalytic triad which is important for the chymotrypsin activity. Predict possible effect of mutations whether active, partially active or inactive based on the property of amino acid residues in the enzyme active site. **Justify** your answer. (Don't just write "active", "partially active" nor "inactive").

a.When Asp102 is mutated to Asn.
b.When Asp102 is mutated to Glu.
c. When Ser195 is mutated to Thr.